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mysql> create database ex1;
Query OK, 1 row affected (0.00 sec)

mysql> use ex1;
Database changed
mysql> CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(50),email VARCHAR(50),role_id INT);
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_name VARCHAR(50));
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name VARCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE customers(customer_id INT PRIMARY KEY,customer_name VARCHAR(50),city VARCHAR(50));
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id INT,order_date DATE,total_amount DECIMAL(10,2));
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_name VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.02 sec)

mysql> -- roles
mysql> INSERT INTO roles (role_id, role_name) VALUES
-> (1, 'Admin'),
-> (2, 'Manager'),
-> (3, 'Employee'),
-> (4, 'HR'),
-> (5, 'Customer Support');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> -- users
mysql> INSERT INTO users (user_id, name, email, role_id) VALUES
-> (101, 'Ravi', 'ravi@gmail.com', 1),
-> (102, 'Priya', 'priya@gmail.com', 2),
-> (103, 'Arjun', 'arjun@gmail.com', 3),
-> (104, 'Sneha', 'sneha@gmail.com', 4),
-> (105, 'Kiran', 'kiran@gmail.com', 3);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> -- departments
mysql> INSERT INTO departments (dept_id, dept_name) VALUES

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mysql> -- departments
mysql> INSERT INTO departments (dept_id, dept_name) VALUES
    -> (10, 'HR'),
    -> (20, 'Finance'),
    -> (30, 'IT'),
    -> (40, 'Sales'),
    -> (50, 'Marketing');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> -- departments
mysql> INSERT INTO departments (dept_id, dept_name) VALUES
    -> (10, 'HR'),
    -> (20, 'Finance'),
    -> (30, 'IT'),
    -> (40, 'Sales'),
    -> (50, 'Marketing');
ERROR 1062 (23000): Duplicate entry '10' for key 'departments.PRIMARY'
mysql> -- employees
mysql> INSERT INTO employees (emp_id, emp_name, dept_id, manager_id, salary) VALUES
    -> (1001, 'Amit', 30, NULL, 90000.00),
    -> (1002, 'Neha', 30, 1001, 60000.00),
    -> (1003, 'Rahul', 20, 1001, 55000.00),
    -> (1004, 'Pooja', 10, 1001, 50000.00),
    -> (1005, 'Vijay', 40, 1001, 65000.00);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> -- customers
mysql> INSERT INTO customers (customer_id, customer_name, city) VALUES
    -> (201, 'Ramesh', 'Hyderabad'),
    -> (202, 'Suresh', 'Bengaluru'),
    -> (203, 'Anjali', 'Chennai'),
    -> (204, 'Meena', 'Mumbai'),
    -> (205, 'Karthik', 'Delhi');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> -- orders
mysql> INSERT INTO orders (order_id, customer_id, order_date, total_amount) VALUES
    -> (5001, 201, '2026-08-01', 2500.00),
    -> (5002, 202, '2026-08-02', 3200.50),
    -> (5003, 203, '2026-08-03', 1500.75),
    -> (5004, 201, '2026-08-04', 4500.00),

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mysql> -- accounts
mysql> INSERT INTO accounts (account_no, holder_name, balance) VALUES
-> (10001, 'Ravi', 50000.00),
-> (10002, 'Priya', 75000.50),
-> (10003, 'Arjun', 30000.00),
-> (10004, 'Sneha', 90000.75),
-> (10005, 'Kiran', 45000.25);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Amit     | IT        |
| Neha     | IT        |
| Rahul    | Finance   |
| Pooja    | HR        |
| Vijay    | Sales     |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email           | role_name |
+-----+-----+-----+
| Ravi  | ravi@gmail.com  | Admin     |
| Priya | priya@gmail.com | Manager   |
| Arjun | arjun@gmail.com | Employee   |
| Sneha | sneha@gmail.com | HR        |
| Kiran | kiran@gmail.com | Employee   |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+
| name | role_name |
+-----+-----+
| Ravi  | Admin     |
| Priya | Manager   |
| Kiran | Employee   |
| Arjun | Employee   |
| Sneha | HR        |
| NULL  | Customer Support |
+-----+-----+
6 rows in set (0.00 sec)
```

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mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
+-----+-----+-----+
| name | email           | role_name |
+-----+-----+-----+
| Ravi  | ravi@gmail.com  | Admin     |
| Priya | priya@gmail.com | Manager   |
| Arjun | arjun@gmail.com | Employee   |
| Sneha | sneha@gmail.com | HR        |
| Kiran | kiran@gmail.com | Employee   |
| NULL  | NULL           | Customer Support |
+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Amit     | NULL    |
| Neha     | Amit    |
| Rahul    | Amit    |
| Pooja    | Amit    |
| Vijay    | Amit    |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
+-----+-----+-----+-----+
| order_id | customer_name | city       | total_amount |
+-----+-----+-----+-----+
| 5001     | Ramesh       | Hyderabad | 2500.00      |
| 5002     | Suresh       | Bengaluru  | 3200.50      |
| 5003     | Anjali       | Chennai   | 1500.75      |
| 5004     | Ramesh       | Hyderabad | 4500.00      |
| 5005     | Karthik      | Delhi     | 1800.00      |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
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mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
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dept_name	COUNT(*)	AVG(salary)	MAX(salary)
IT	2	75000.000000	90000.00
Finance	1	55000.000000	55000.00
HR	1	50000.000000	50000.00
Sales	1	65000.000000	65000.00

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4 rows in set (0.00 sec)

mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';
Empty set (0.00 sec)

mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
```

customer_name	TotalPurchase
Ramesh	7000.00
Suresh	3200.50
Karthik	1800.00
Anjali	1500.75

```
4 rows in set (0.00 sec)

mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
ERROR 1146 (42502): Table 'ex1.students' doesn't exist
mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000); COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000); ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=101; UPDATE accounts SET balance=balance+5000 WHERE account_no=102; COMMIT;
Query OK, 0 rows affected (0.00 sec)
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Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;
Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.00 sec)
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